

Module 13

Functional & Mechanical Design — Tier 3: Advanced

1. Designing Joints

- **Snap-fit joints** — no hardware needed, relies on plastic flex
- **Press-fit joints** — precise tolerance holds parts together by friction
- **Hardware joints** — designed around actual screws/bolts/nuts (heat-set inserts are common for durability)

2. Tolerances for Moving Parts

Parts that need to move (hinges, sliders) typically need 0.3–0.5mm clearance to avoid fusing together or binding due to print imprecision.

3. Multi-Part Assemblies

- Design alignment features (pins, notches) so parts can only fit one correct way
- Print orientation matters even more for assemblies — weak print directions can fail under assembly stress
- Test-fit with a quick low-quality print before committing to a final high-quality print

End of Module 13 — Next: Module 14, Production & Scaling